

Beam Report

Beam A-1

Beam Name	A-1
Span (ft)	12.0
Load (lb)	4000.0
FOS	1.5
Deflection Divisor	360.0
Span Inches	144.00
Allowed Delta	0.4000
Max Bend	9.00kip-ft
Required Shear	13.41In^4
Max Shear	20.11In^4

Recommended Beams

Recommended Beam	Ix	Moment Capacity
W5X16	21.00	23.00 kip-ft
W6X12	22.00	20.00 kip-ft
W5X19	26.00	27.00 kip-ft
W6X15	29.00	26.00 kip-ft
*W8X10	31.00	21.00 kip-ft

Calculation Steps

Span: $12.0 \times 12 = 144.00$ in

Delta: $144.00 \div 360.0 = 0.4000$ in

Max Bend: $((4000.0 \times 12.0) \div 8) \times 1.5 \div 1000 = 9.00$ ft-lb

Required Shear: $(5 \times 4000.0 \times 144.00^3) \div (384 \times 29,000,000 \times 0.4000) = 13.41$

Max Shear: $13.41 \times 1.5 = 20.11$

Beam A-2

Beam Name	A-2
Span (ft)	16.0
Load (lb)	8000.0
FOS	1.5
Deflection Divisor	360.0
Span Inches	192.00
Allowed Delta	0.5333
Max Bend	24.00kip-ft
Required Shear	47.67In^4
Max Shear	71.50In^4

Recommended Beams

Recommended Beam	Ix	Moment Capacity
W8X21	75.00	48.00 kip-ft
W10X17	82.00	42.00 kip-ft
W8X24	83.00	56.00 kip-ft
*W12X14	89.00	39.00 kip-ft
W10X19	96.00	50.00 kip-ft

Calculation Steps

Span: $16.0 \times 12 = 192.00$ in

Delta: $192.00 \div 360.0 = 0.5333$ in

Max Bend: $((8000.0 \times 16.0) \div 8) \times 1.5 \div 1000 = 24.00$ ft-lb

Required Shear: $(5 \times 8000.0 \times 192.00^3) \div (384 \times 29,000,000 \times 0.5333) = 47.67$

Max Shear: $47.67 \times 1.5 = 71.50$